

Class 09: Candy Mini-Project

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Background

In this mini-project, you will explore FiveThirtyEight's Halloween Candy dataset.

Today we are taking a detour to analyze a fun dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA).

Importing candy data

The data is all related to halloween candy and is from the 538 website.

```
candy_file <- "candy-data.txt"

candy = read.csv(candy_file, row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0
	hard bar	pluribus	sugarpercent	pricepercent	winpercent	
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

What is in the dataset?

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

There are 85 different candy types.

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

There are 38 fruity candy types.

What is your favorite candy?

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["3 Musketeers", ]$winpercent
```

```
[1] 67.60294
```

My favorite candy is 3 Musketeers and its winpercent value is 67.60294.

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

The winpercent value for Kit Kat is 76.7686.

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

The winpercent value for Tootsie Roll Snack Bars is 49.6535.

There is a useful `skim()` package.

```
# install.packages("skimr")  
library("skimr")  
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency: numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The sugarpercent looks to be on a different scale as it represents the percentile of the sugar content, not a value of 0 or 1.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

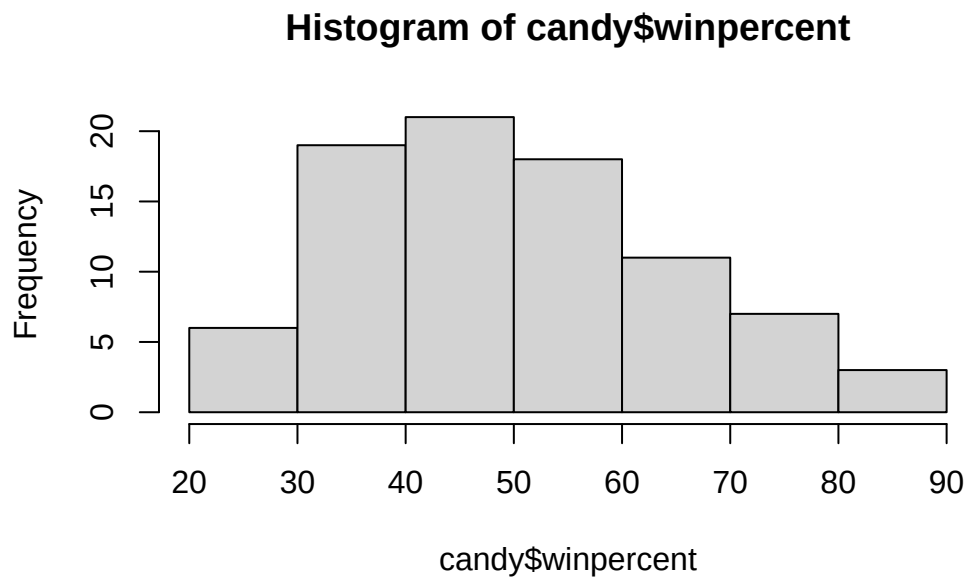
A zero represents that the candy is not a chocolate, whereas a one represents that the candy is a chocolate.

Exploratory analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

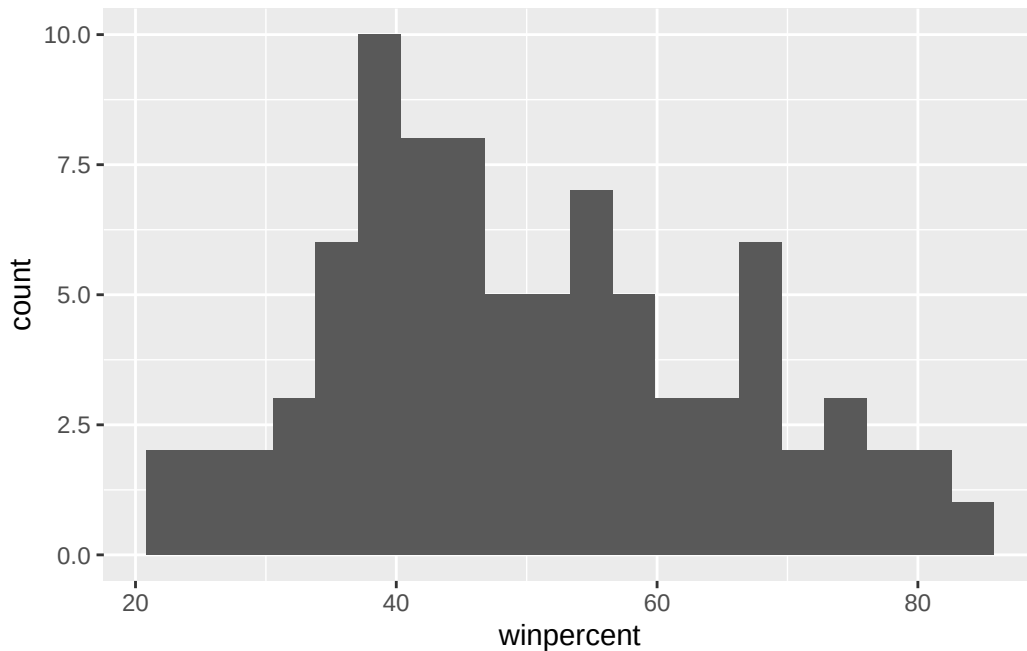
Histogram using base R:

```
hist(candy$winpercent)
```



Histogram using ggplot2:

```
library(ggplot2)
ggplot(candy, aes(winpercent)) +
  geom_histogram(bins = 20)
```



Q9. Is the distribution of winpercent values symmetrical?

No, the distribution of winpercent values is not symmetrical.

Q10. Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

If we calculate by mean, the center of the distribution is slightly above 50%. However, if we calculate by median, which seems to be more appropriate based on the histogram, the center of the distribution is slightly below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[as.logical(candy$fruit)])
```

```
[1] 44.11974
```

```
## choc.ind <- as.logical(candy$chocolate)
## choc.candy <- candy[choc.ind, ]
## choc.win <- choc.candy$winpercent
## mean(choc.win)
```

On average, chocolate candy is higher ranked than fruit candy.

Q12. Is this difference statistically significant?

```
t.test(
  candy$winpercent[as.logical(candy$chocolate)],
  candy$winpercent[as.logical(candy$fruit)]
)
```

Welch Two Sample t-test

```
data: candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fruit)]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

This difference is statistically significant as the p-value is less than 0.05.

Overall candy rankings

Q13. What are the five least liked candy types in this set?

```
head(candy[ order(candy$winpercent), ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197		0.976
Boston Baked Beans				0	0	0	1	0.313		0.511
Chiclets				0	0	0	1	0.046		0.325
Super Bubble				0	0	0	0	0.162		0.116
Jawbusters				0	1	0	1	0.093		0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

The five least liked candies are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

```
tail(candy[ order(candy$winpercent), ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Snickers	1	0	1		1	1
Kit Kat	1	0	0		0	0
Twix	1	0	1		0	0
Reese's Miniatures	1	0	0		1	0
Reese's Peanut Butter cup	1	0	0		1	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Snickers				0	0	1	0	0.546
Kit Kat				1	0	1	0	0.313
Twix				1	0	1	0	0.546
Reese's Miniatures				0	0	0	0	0.034
Reese's Peanut Butter cup				0	0	0	0	0.720

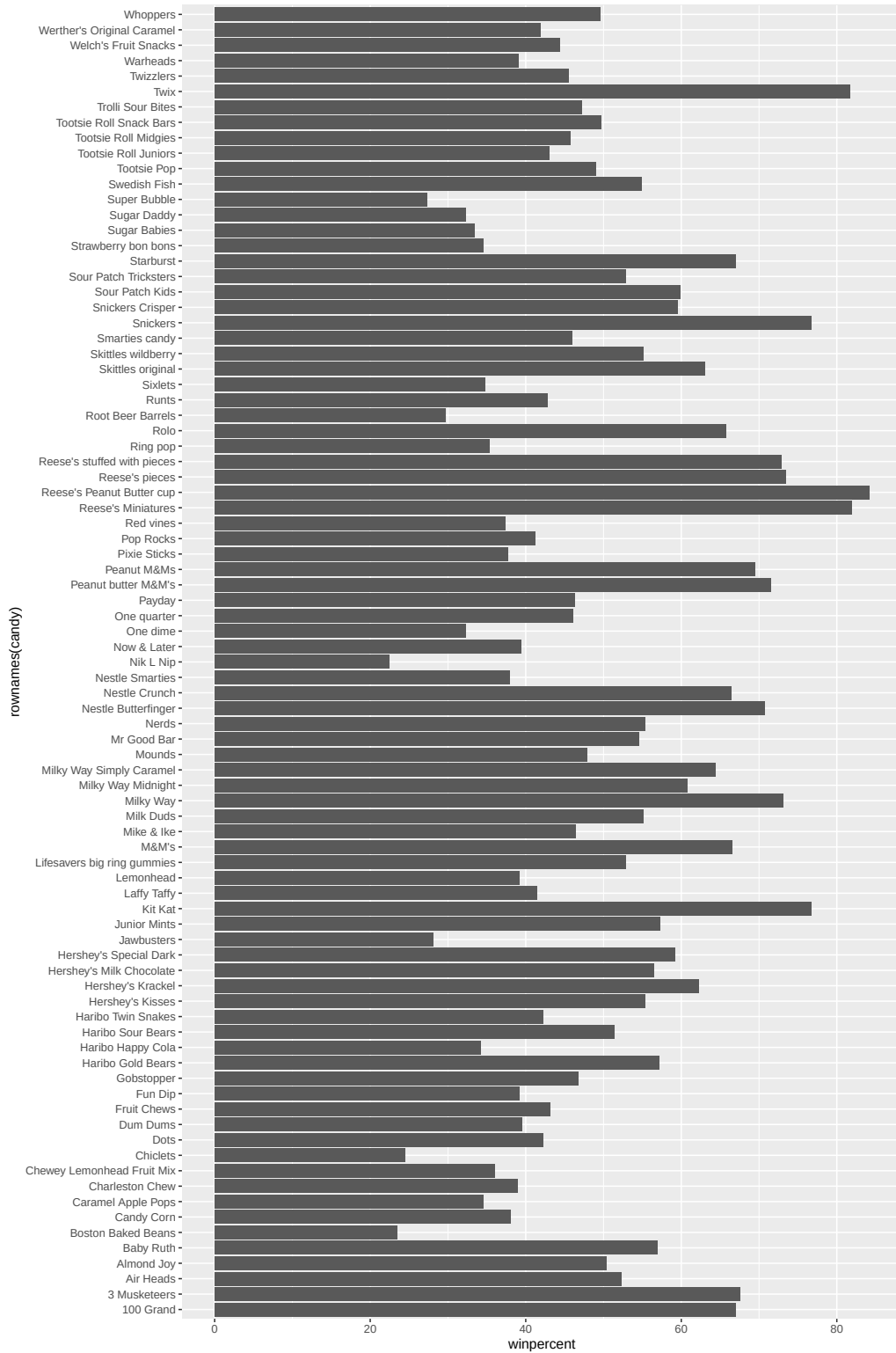
	price	percent	winpercent
Snickers	0.651		76.67378
Kit Kat	0.511		76.76860
Twix	0.906		81.64291

Reese's Miniatures	0.279	81.86626
Reese's Peanut Butter cup	0.651	84.18029

The top 5 all time favorite candy types out of this set are Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

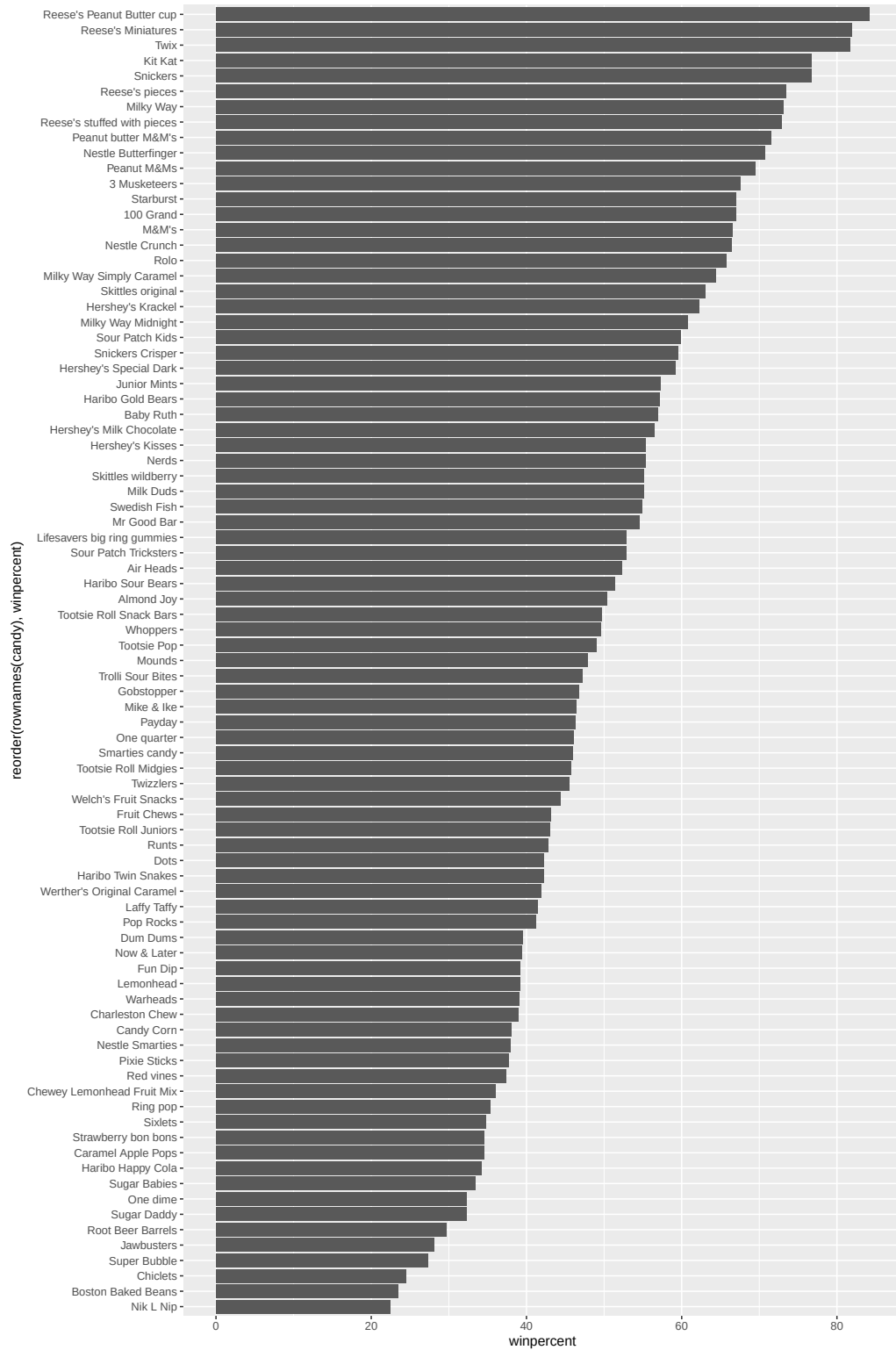
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

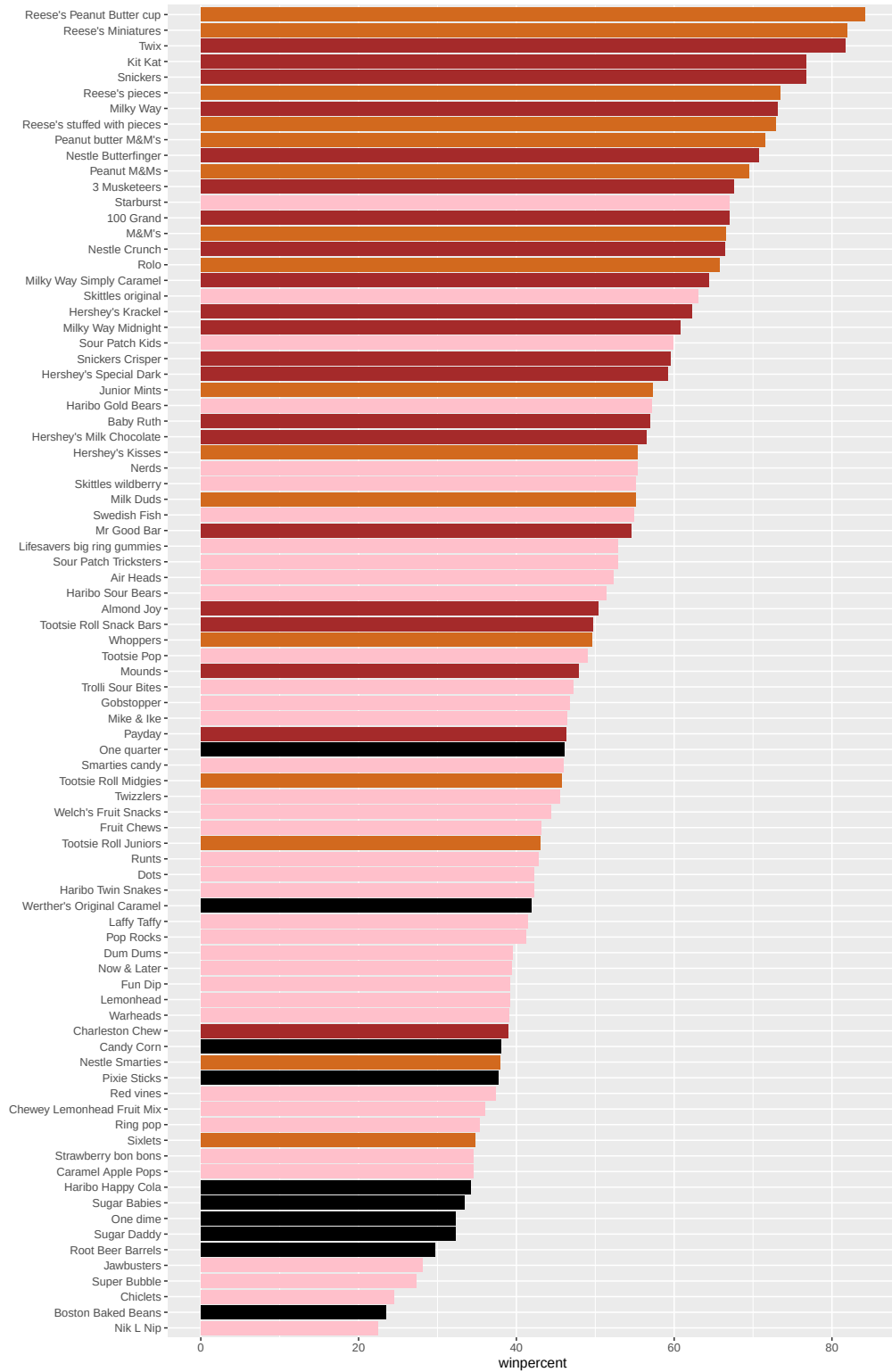
```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +  
  geom_col()
```



Time to add some useful color

```
my_cols = rep("black", nrow(candy))
## chocolate candy in chocolate color
my_cols[as.logical(candy$chocolate)] = "chocolate"
## candy bars in brown
my_cols[as.logical(candy$bar)] = "brown"
## fruity candy in pink
my_cols[as.logical(candy$fruity)] = "pink"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```



Q17. What is the worst ranked chocolate candy?

Sixlets is the worst ranked chocolate candy.

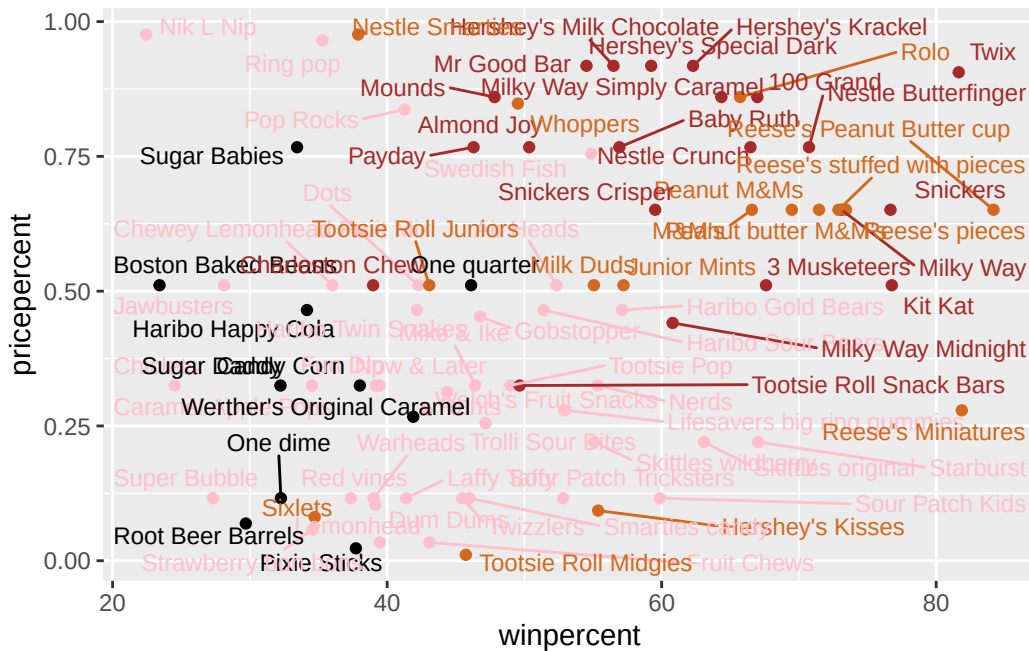
Q18. What is the best ranked fruity candy?

Starbursts is the best ranked fruity candy.

Taking a look at pricepercent

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(x = winpercent, y = pricepercent, label = rownames(candy)) +
  geom_point(col = my_cols) +
  geom_text_repel(col = my_cols, size = 3.3, max.overlaps = 25)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
head(candy[order(-candy$winpercent / candy$pricepercent), ], 5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Tootsie Roll Midgies	1	0	0	0	0
Pixie Sticks	0	0	0	0	0
Fruit Chews	0	1	0	0	0
Dum Dums	0	1	0	0	0
Strawberry bon bons	0	1	0	0	0

	crispedricewafer	hard bar	pluribus	sugarpercent
Tootsie Roll Midgies	0	0	1	0.174
Pixie Sticks	0	0	1	0.093
Fruit Chews	0	0	1	0.127
Dum Dums	0	1	0	0.732
Strawberry bon bons	0	1	1	0.569

	pricepercent	winpercent
Tootsie Roll Midgies	0.011	45.73675
Pixie Sticks	0.023	37.72234
Fruit Chews	0.034	43.08892
Dum Dums	0.034	39.46056
Strawberry bon bons	0.058	34.57899

Tootsie Roll Midgies is the highest ranked in terms of winpercent for the least money.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
top5 <- head( candy[ord,c(11,12)], n=5 )

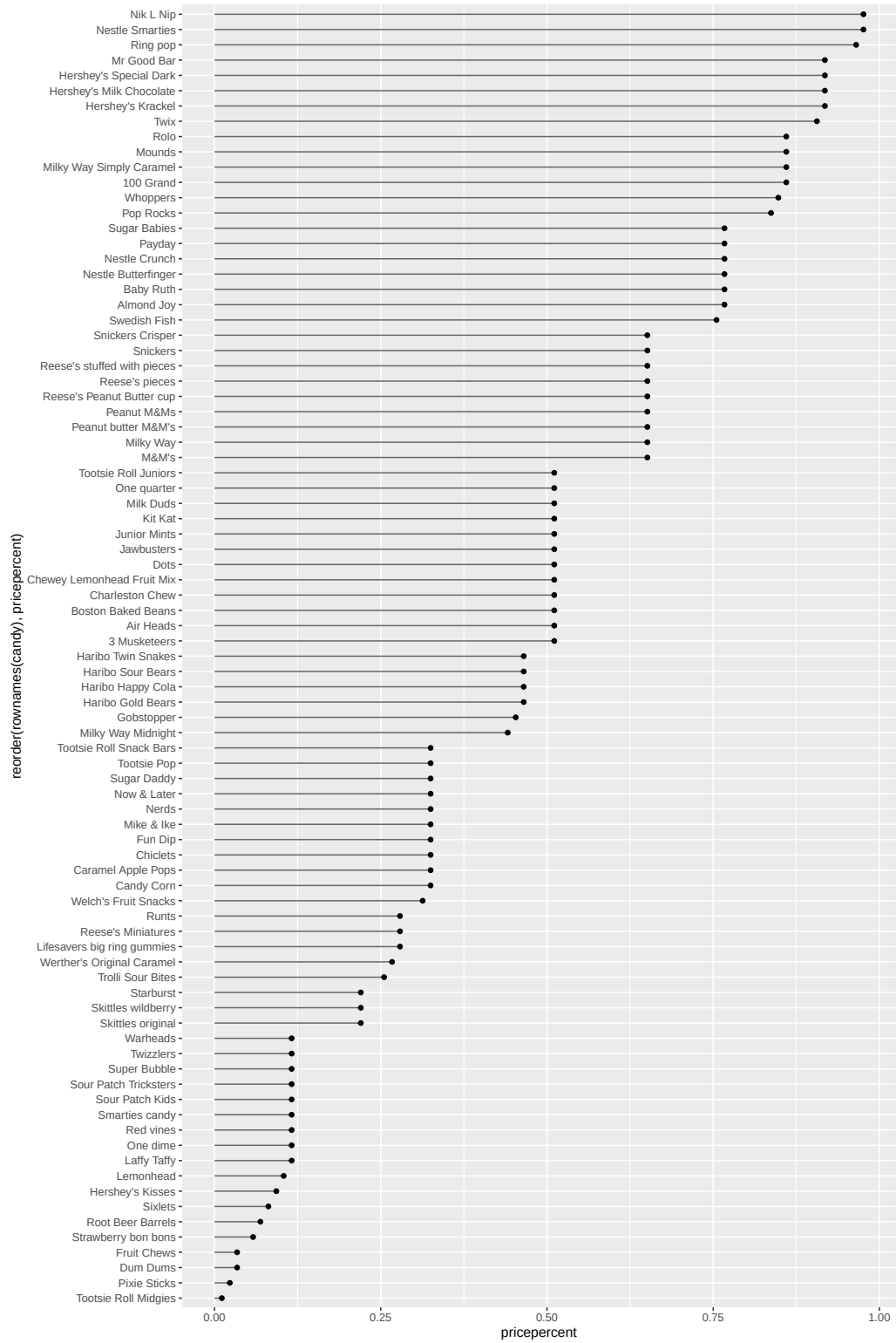
top5[order(top5$winpercent), ]
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Ring pop	0.965	35.29076
Nestle Smarties	0.976	37.88719
Hershey's Milk Chocolate	0.918	56.49050
Hershey's Krackel	0.918	62.28448

The least popular of the top 5 most expensive candy types in the dataset are Nik L Nip, Ring pop, Nestle Smarties, Hershey's Milk Chocolate, and Hershey's Krackel.

Optional

```
ggplot(candy) +  
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +  
  geom_segment(aes(yend = reorder(rownames(candy), pricepercent),  
                  xend = 0), col="gray40") +  
  geom_point()
```

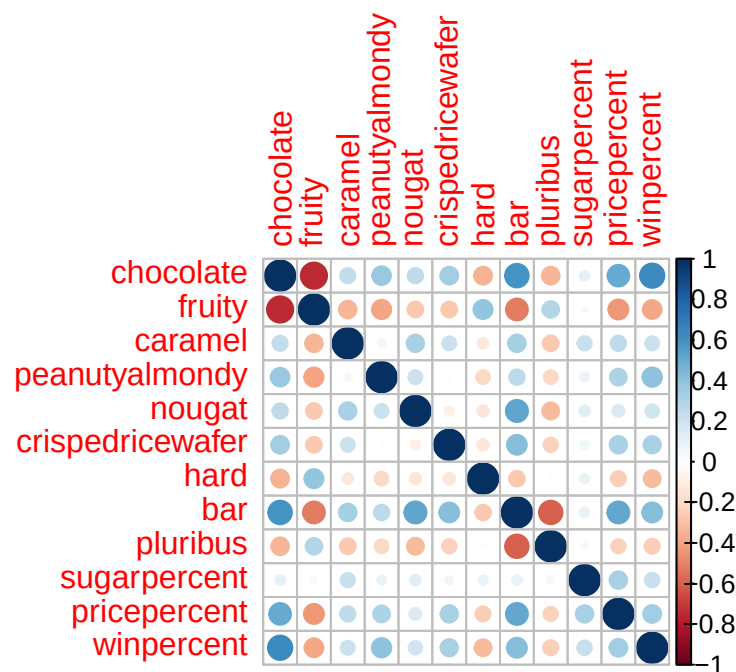


Exploring the correlation structure

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)  
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruity are strongly anti-correlated. This makes sense because candies are typically one or the other, not both.

Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

I expect chocolate, bar, and peanutyalmondy to have strong positive loadings, and fruity, hard, and pluribus to have strong negative loadings on PC1, showing a major separation between chocolate candies and fruity candies.

Principal Component Analysis

In this case we want to be sure to set `scale = TRUE` argument for `prcomp()` because we have one variable, `winpercent`, that is on a very different scale than all others and would otherwise dominate our PCA results.

```
candy_file <- "candy-data.txt"
candy = read.csv(candy_file, row.names = 1)
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

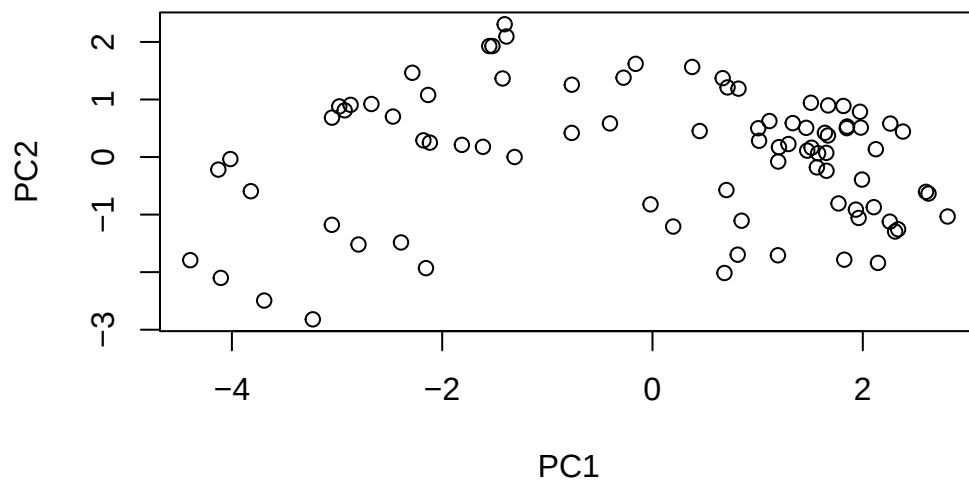
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

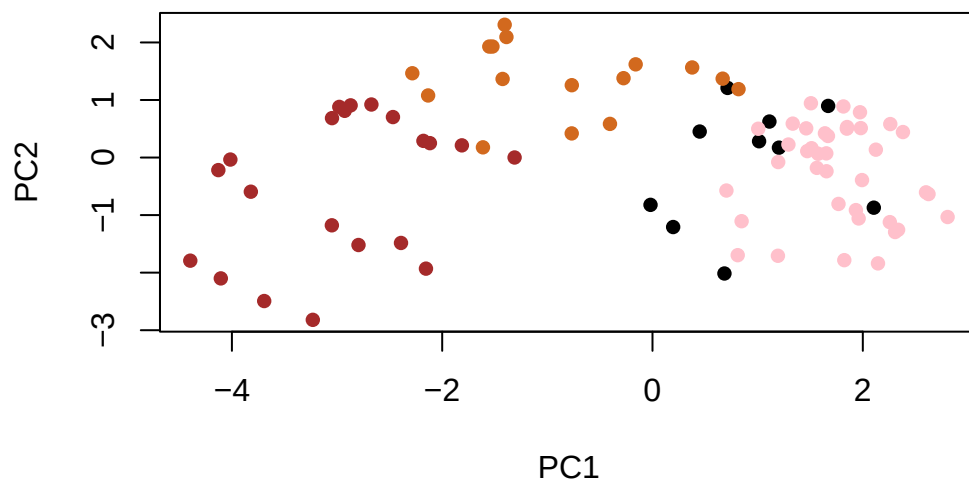
	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

First major result figure is the “score plot” of PC1 vs PC2 - how the different candies are related to each other on our new PC axis:

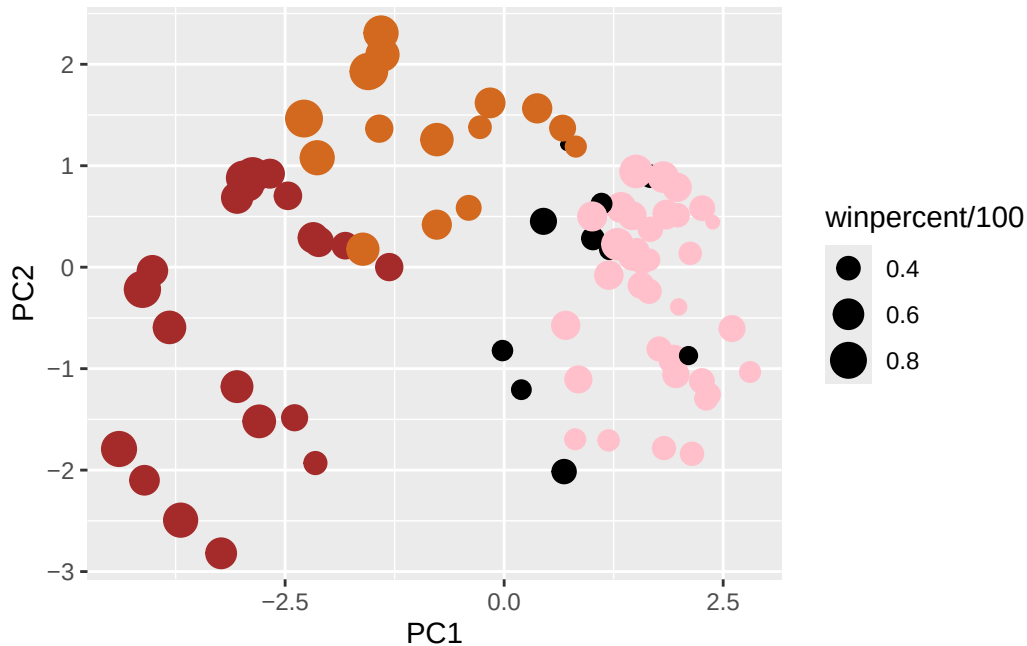
```
plot(pca$x[,1:2])
```



```
plot(pca$x[,1:2], col=my_cols, pch=16)
```



```
my_data <- cbind(candy, pca$x[,1:3])
p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,
      size = winpercent/100,
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = my_cols)
p
```

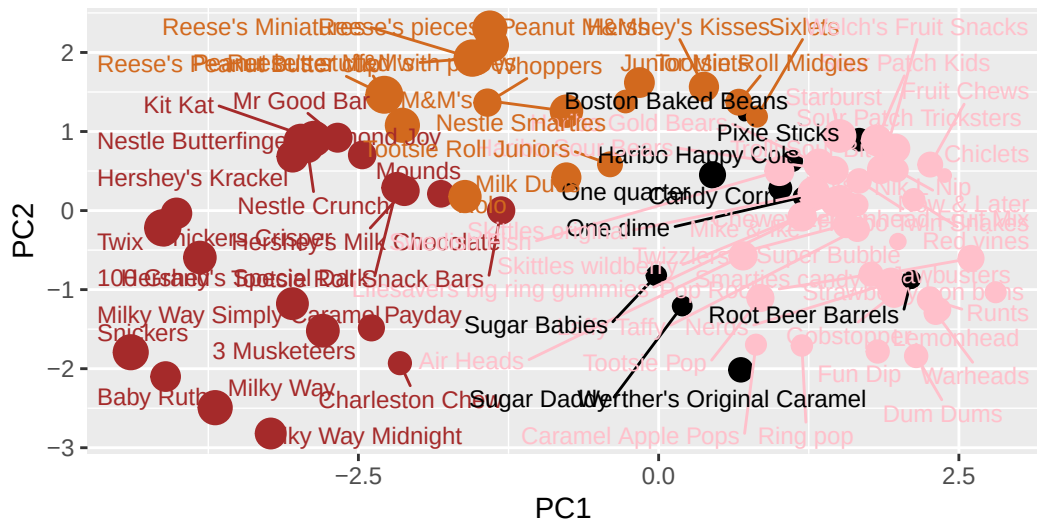


```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 40) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
       caption="Data from 538")
```

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



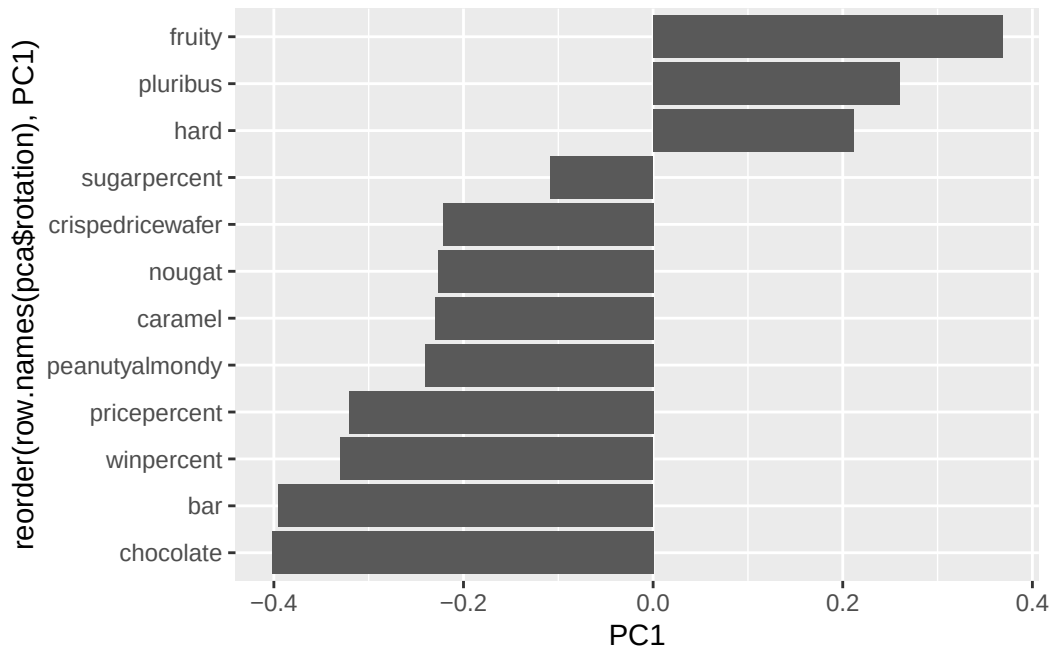
Data from 538

```
## library(plotly)
## ggplotly(p)
```

The second major results figure from PCA is the so-called “loadings plot”.

Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

```
ggplot(pca$rotation) +
  aes(x = PC1,
       reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



The variables `fruity`, `pluribus`, and `hard` were picked up strongly by PC1 in the positive direction. This makes sense as this relationship was previously highlighted when looking at the correlation structure graph.

Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A “winning” candy in this dataset is typically chocolate, especially in bar form, frequently with ingredients such as peanuts, caramel, or nougat. These candies tend to have higher `winpercent` values, meaning they are more consistently preferred. The exploratory analysis supports this by showing that chocolate candies have higher average `winpercent` than fruit ones. In addition, the correlation analysis further emphasizes that chocolate is positively associated with `winpercent` as well as `bar` and `peanutyalmondy`, while `fruity` has a negative correlation. Finally, the PCA results shows that the PC1 separates candies along a chocolate vs fruity axis. Variables including `chocolate`, `bar`, and `peanutyalmondy` move strongly in the positive direction, while `fruity` and `hard` go in the opposite direction.